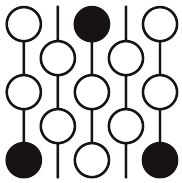
Review of Problem 5 – Humanize



First Proof Third Batch
Reviewer 1
October 8–9, 2026

1 Review

This submission does not resolve the question. It only claims to prove some partial results. It verifies the claim in some special cases. The statement is false in general, and the solution make no progress toward a counterexample.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: -0.5

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: -1

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 0

2 Submission with inline comments

Abstract

Problem. Let $\mathbb{H}$ be a hyperfield with hyperaddition $\boxplus$. For $x, y, z \in \mathbb{H}$ with $0 \in x \boxplus y \boxplus z$, does it follow that $0 \in x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz)$?

Content of this document. We give complete proofs, from the hyperfield axioms alone, of the degenerate case $xyz = 0$ (Proposition 4.3) and of the case in which one of the three pair sums $x \boxplus y$, $x \boxplus z$, $y \boxplus z$ is a singleton (Corollary 5.2). The remaining case is isolated in the precisely stated Elevation Problem 6.4, which is equivalent to the unrestricted statement; the document proves the conditional implication (Elevation Problem $\Rightarrow$ full theorem) in Theorem 4.1. We do not assert the unrestricted answer "yes" as proved in this round; no computation, and no external theorem or unproved lemma is used as a proof step anywhere below.

3 Preliminaries

Definition 3.1. A *hyperfield* is a set $\mathbb{H}$ with an element 0, a commutative multiplication making $\mathbb{H}^\times := \mathbb{H} \setminus \{0\}$ an abelian group (with $0 \cdot a = 0$), the abbreviation $-a := (-1)a$, and a hyperaddition $\boxplus : \mathbb{H} \times \mathbb{H} \to \mathcal{P}(\mathbb{H}) \setminus \{\varnothing\}$ satisfying:

(H1) $x \boxplus y = y \boxplus x$; $x \boxplus 0 = \{x\}$; $0 \in x \boxplus y \iff y = -x$;

(H2) $(x \boxplus y) \boxplus z = x \boxplus (y \boxplus z)$, where for $S \subseteq \mathbb{H}$ one sets $S \boxplus z := \bigcup_{s \in S} s \boxplus z$. All bracketings of an iterated sum coincide; we write $x_1 \boxplus \cdots \boxplus x_n$, and $a^{\boxplus k}$ for the k-fold hypersum of a with itself;

(H3) $a(x \boxplus y) \subseteq ax \boxplus ay$;

(H4) $t \in x \boxplus y \implies x \in t \boxplus (-y)$.

Lemma 3.2 (elementary consequences)**.** *Let $a, x, y, z \in \mathbb{H}$, $a \neq 0$.*

(i) $a(x \boxplus y) = ax \boxplus ay$;

(ii) $-(x \boxplus y) = (-x) \boxplus (-y)$;

(iii) $0 \in x \boxplus y \boxplus z \iff -z \in x \boxplus y$;

(iv) *if $x \boxplus y = \{t\}$ is a singleton with $x, y \neq 0$, then $-y \in x \boxplus (-t)$ and $-x \in y \boxplus (-t)$;*

(v) *if $0 \in x \boxplus y \boxplus z$ then $0 \in (ax) \boxplus (ay) \boxplus (az)$.*

Proof. (i) (H3) gives $a(x \boxplus y) \subseteq ax \boxplus ay$; applying (H3) with scalar a^{-1} gives $a^{-1}(ax \boxplus ay) \subseteq x \boxplus y$, hence the reverse inclusion $ax \boxplus ay \subseteq a(x \boxplus y)$. (ii) is (i) with $a = -1$. (iii) If $0 \in x \boxplus y \boxplus z$, pick $t \in x \boxplus y$ with $0 \in t \boxplus z$; by (H1), $z = -t \in -(x \boxplus y)$, so $-z \in x \boxplus y$ by (ii). Conversely, if $-z \in x \boxplus y$ then $0 \in (-z) \boxplus z \subseteq x \boxplus y \boxplus z$ by (H1). (iv) We have $t \in x \boxplus y$. By (H4), $x \in t \boxplus (-y)$; then $0 \in x \boxplus (-x) \subseteq (t \boxplus (-y)) \boxplus (-x)$ by (H1). The hypersum $(t \boxplus (-y)) \boxplus (-x)$ contains 0, so by (iii)

$$-t \;\in\; (-y) \boxplus (-x) \;=\; -(x \boxplus y) \;=\; \{-t\},$$

the middle equality being (ii). Hence $-t \in (-x) \boxplus (-y)$; applying (H4) to this membership gives $-y \in x \boxplus (-t)$. The proof of $-x \in y \boxplus (-t)$ is symmetric, exchanging the roles of x and y. (v) From $-z \in x \boxplus y$ (iii) and (i) we get $-az \in ax \boxplus ay$, and (iii) again gives the claim. □

For a finite multiset M we write $\boxplus M$ for the (unambiguous) hypersum of its elements.

Lemma 3.3 (substitution atoms)**.** *Let $a, b, c \in \mathbb{H}^\times$ with $-c \in a \boxplus b$.*

(A1) *$0 \in a \boxplus b \boxplus c$ and, for every $k \geq 1$, $0 \in a^{\boxplus k} \boxplus b^{\boxplus k} \boxplus c^{\boxplus k}$.*

(A2) (cubing a pair sum) *With $(a \boxplus b)^{\cdot 3} := \{t_1 t_2 t_3 : t_i \in a \boxplus b\}$,*

$$-c^3 \;\in\; (a \boxplus b)^{\cdot 3} \;\subseteq\; a^3 \boxplus \left(a^2 b\right)^{\boxplus 3} \boxplus \left(ab^2\right)^{\boxplus 3} \boxplus b^3.$$

(A3) *For every $\lambda \in \mathbb{H}^\times$, $-\lambda c \in \lambda a \boxplus \lambda b$.*

Proof. (A1) The case $k = 1$ is Lemma 3.2(iii). For general k, use the same value $-c \in a \boxplus b$ in each of k blocks: by (H2) the $3k$-term hypersum obtained from k disjoint blocks each isomorphic to $a \boxplus b \boxplus c$ contains 0 (realize each block by the value $-c$ and each pair of block-outputs by 0); by (H1) this multiset is exactly $a^{\boxplus k} \cup b^{\boxplus k} \cup c^{\boxplus k}$.

(A2) Choosing $t_1 = t_2 = t_3 = -c$ gives $-c^3 \in (a \boxplus b)^{\cdot 3}$. For the inclusion, let $u, v \in a \boxplus b$. By Lemma 3.2(i),

$$uv \;\in\; u(a \boxplus b) \;=\; ua \boxplus ub \;\subseteq\; \left(a^2 \boxplus ab\right) \boxplus \left(ab \boxplus b^2\right) \;=\; a^2 \boxplus (ab)^{\boxplus 2} \boxplus b^2,$$

using $u \in a \boxplus b$ to get $ua \in a^2 \boxplus ab$ and $ub \in ab \boxplus b^2$. Now any $t = t_1 t_2 t_3 \in (a \boxplus b)^{\cdot 3}$ is $(t_1 t_2) t_3$ with $u := t_1 t_2 \in a^2 \boxplus (ab)^{\boxplus 2} \boxplus b^2$ and $t_3 \in a \boxplus b$, whence

$$t = u t_3 \;\in\; \left(a^2 \boxplus (ab)^{\boxplus 2} \boxplus b^2\right) t_3 \;\subseteq\; a^3 \boxplus (a^2 b)^{\boxplus 3} \boxplus (ab^2)^{\boxplus 3} \boxplus b^3,$$

since $\gamma t_3 \in \gamma a \boxplus \gamma b$ for each $\gamma \in \{a^2, ab, b^2\}$ (Lemma 3.2(i)), and (H1) collects the exponents.

(A3) is Lemma 3.2(i) applied to $-c \in a \boxplus b$. □

4 Statement and degenerate case

Theorem 4.1 (conditional form)**.** *Let $\mathbb{H}$ be a hyperfield and $x, y, z \in \mathbb{H}$ with $0 \in x \boxplus y \boxplus z$. Then*

$$0 \;\in\; \Sigma \;:=\; x^3 \boxplus y^3 \boxplus z^3 \boxplus (-w) \boxplus (-w) \boxplus (-w), \qquad w := xyz,$$

provided the Elevation Problem 6.4 below has an affirmative answer. The degenerate instance $w = 0$ and the collapse instance (some pair sum among $\{x, y, z\}$ a singleton) are unconditional (Proposition 4.3, Corollary 5.2).

Lemma 4.2 (scalar invariance)**.** *For $\lambda \in \mathbb{H}^\times$, the implication $0 \in x \boxplus y \boxplus z \Rightarrow 0 \in \Sigma$ holds for (x, y, z) iff it holds for $(\lambda x, \lambda y, \lambda z)$.*

Proof. The hypothesis is invariant by Lemma 3.2(v). The conclusion hypersum of the scaled triple equals $\lambda^3 \Sigma$ by repeated use of Lemma 3.2(i), and $0 \in \lambda^3 \Sigma \iff 0 \in \Sigma$. □

Proposition 4.3 (degenerate case)**.** *If $xyz = 0$ and $0 \in x \boxplus y \boxplus z$, then $0 \in \Sigma$.*

Proof. Permuting the variables, we may assume $z = 0$. For t ranging over $x \boxplus y$, (H1) gives $t \boxplus z = \{t\}$; since $0 \in x \boxplus y \boxplus z$ some such hypersum $t \boxplus z$ contains 0, so $t = 0$, and (H1) gives $y = -x$; then $w = 0$. If $x = 0$ as well, all six terms of Σ are 0, and any hypersum of zeros is $\{0\}$. Otherwise $y = -x \neq 0$: then $y^3 = (-x)^3 = -x^3$, and $z^3 = 0$, $-w = 0$, so writing the hypersum in the bracketing $\left(x^3 \boxplus y^3\right) \boxplus z^3 \boxplus (-w) \boxplus (-w) \boxplus (-w)$, the first factor contains 0 by (H1) (since $y^3 = -x^3$), and appending the remaining zero terms preserves the value 0 (H1 again: $0 \boxplus 0 = \{0\}$). Hence $0 \in \Sigma$. □

From now on $w = xyz \neq 0$.

5 The collapse case

Lemma 5.1 (collapse lemma)**.** *Let $X, Y \in \mathbb{H}^\times$ with $X \boxplus Y = \{1\}$. Then $0 \in X^3 \boxplus Y^3 \boxplus (-1) \boxplus (XY) \boxplus (XY) \boxplus (XY)$.*

Proof. Set $\mathcal{U} := (X^2Y)^{\boxplus 3} \boxplus (XY^2)^{\boxplus 3}$ and $\mathcal{W} := (XY)^{\boxplus 3}$. Scaling $X \boxplus Y = \{1\}$ by the scalar XY (Lemma 3.2(i)):

$$X^2Y \boxplus XY^2 = \{XY\}.$$

Regrouping the six terms of $\mathcal{U}$ into three such pairs (H1),(H2):

$$\mathcal{U} \;=\; \boxplus_{i=1}^{3} \big(X^2Y \boxplus XY^2\big) \;=\; \{XY\} \boxplus \{XY\} \boxplus \{XY\} \;=\; \mathcal{W}. \tag{1}$$

Apply (A2) to $1 \in X \boxplus Y$ (so $c = -1$ and $-c^3 = 1$):

$$1 \;\in\; (X \boxplus Y)^{\cdot 3} \;\subseteq\; X^3 \boxplus (X^2Y)^{\boxplus 3} \boxplus (XY^2)^{\boxplus 3} \boxplus Y^3,$$

so there exist $g \in X^3 \boxplus Y^3$ and $u \in \mathcal{U}$ with $1 \in g \boxplus u$; by (H1), $0 \in g \boxplus u \boxplus (-1)$, i.e.

$$0 \;\in\; g \boxplus u \boxplus (-1) \qquad \text{for some } g \in X^3 \boxplus Y^3,\; u \in \mathcal{U}. \tag{2}$$

By (1) the value u is an element of $\mathcal{W} = (XY)^{\boxplus 3}$, so the singleton $\{u\}$ is included in the value set of the three-term hypersum $(XY) \boxplus (XY) \boxplus (XY)$; value substitution (H2) in (2) therefore gives

$$0 \;\in\; g \boxplus (-1) \boxplus (XY) \boxplus (XY) \boxplus (XY) \;\subseteq\; X^3 \boxplus Y^3 \boxplus (-1) \boxplus (XY)^{\boxplus 3}. \qquad \square$$

Corollary 5.2 (collapse case of Theorem 4.1)**.** *Let $0 \in x \boxplus y \boxplus z$ with $w = xyz \neq 0$, and suppose some pair sum among $\{x, y, z\}$ is a singleton. Then $0 \in \Sigma$.*

Proof. Permuting the variables, assume $x \boxplus y$ is a singleton. Since $-z \in x \boxplus y$ (Lemma 3.2(iii)), that singleton is $\{-z\}$. Rescale by $\lambda := -z^{-1}$ (Lemma 4.2) and put $X := -z^{-1}x$, $Y := -z^{-1}y$: the scaled triple is $(X, Y, -1)$ with $X \boxplus Y = \{1\}$ (Lemma 3.2(i)), and the scaled value of w is $(-z^{-1})xyz = -XY$. By Lemma 4.2 the conclusion for the scaled triple, namely $0 \in X^3 \boxplus Y^3 \boxplus (-1) \boxplus (XY) \boxplus (XY) \boxplus (XY)$, is exactly Lemma 5.1, and it pulls back to $0 \in \Sigma$ by Lemma 4.2 once more. $\square$

6 The free case: towers and the elevation problem

Assume throughout:

$$0 \in x \boxplus y \boxplus z, \qquad w = xyz \neq 0, \qquad x \boxplus y \neq \{-z\},\; x \boxplus z \neq \{-y\},\; y \boxplus z \neq \{-x\}. \tag{F}$$

Set

$$P := x^2y \boxplus xy^2, \qquad H := x^3 \boxplus y^3 \boxplus z^3, \qquad \overline{V} := (-w) \boxplus (-w) \boxplus (-w).$$

Scaling $-z \in x \boxplus y$ by xy (Lemma 3.2(i)) gives $-w \in P$. The goal is $0 \in H \boxplus \overline{V}$.

Lemma 6.1 (the two towers)**.** *Under* (F)*:*

(i) *$0 \in H \boxplus P \boxplus P \boxplus P$, i.e. there exist $h \in H$ and $p_1, p_2, p_3 \in P$ with $0 \in h \boxplus p_1 \boxplus p_2 \boxplus p_3$;*

(ii) *$0 \in P \boxplus P \boxplus P \boxplus w \boxplus w \boxplus w$.*

Proof. (i) Apply (A2) to $-z \in x \boxplus y$: $-z^3 \in x^3 \boxplus (x^2y)^{\boxplus 3} \boxplus (xy^2)^{\boxplus 3} \boxplus y^3$, whence by Lemma 3.2(iii) $0 \in H \boxplus (x^2y)^{\boxplus 3} \boxplus (xy^2)^{\boxplus 3}$ after regrouping (H1). The six mixed terms regroup into three pairs (H1),(H2):

$$(x^2y)^{\boxplus 3} \boxplus (xy^2)^{\boxplus 3} \;=\; \boxplus_{i=1}^{3} \big(x^2y \boxplus xy^2\big) \;=\; P \boxplus P \boxplus P, \tag{3}$$

which gives (i). (ii) Apply (A1) with $k = 3$ to $-w \in P$: the blocks are the relation $-w \in x^2y \boxplus xy^2$, i.e. $0 \in x^2y \boxplus xy^2 \boxplus w$, three times; their union regroups (H1) into $P \boxplus P \boxplus P \boxplus w^{\boxplus 3}$. □

Lemma 6.2 (free propagation)**.** *Under* (F)*, each of the following pair sums has at least two elements:*

$$x^2y \boxplus xy^2, \qquad x^2z \boxplus xz^2, \qquad y^2z \boxplus yz^2, \qquad C \boxplus w,\;\; C \boxplus (-w) \;\; \text{for} \;\; C \in \{x^2y, xy^2, x^2z, xz^2, y^2z, yz^2\}.$$

Proof. Each displayed pair sum is, by Lemma 3.2(i), a scalar multiple of one of the level-1 pair sums $x \boxplus y$, $x \boxplus z$, $y \boxplus z$: with $w = xyz$,

$$\begin{aligned} x^2y \boxplus xy^2 &= xy\,(x \boxplus y), & x^2z \boxplus xz^2 &= xz\,(x \boxplus z), & y^2z \boxplus yz^2 &= yz\,(y \boxplus z),\\ x^2y \boxplus w &= xy\,(x \boxplus z), & xy^2 \boxplus w &= xy\,(y \boxplus z), & x^2z \boxplus w &= xz\,(x \boxplus y),\\ xz^2 \boxplus w &= xz\,(y \boxplus z), & y^2z \boxplus w &= yz\,(x \boxplus y), & yz^2 \boxplus w &= yz\,(x \boxplus z), \end{aligned}$$

and by Lemma 3.2(ii) each $C \boxplus (-w)$ is the negative of the corresponding $C \boxplus w$. Scalar multiples preserve the cardinality of a pair sum. Finally, each of $x \boxplus y$, $x \boxplus z$, $y \boxplus z$ has at least two elements: $-z \in x \boxplus y$, so if $x \boxplus y$ were a singleton it would equal $\{-z\}$, excluded by (F); similarly $-y \in x \boxplus z$ and $-x \in y \boxplus z$ exclude the other two. □

Remark 6.3 (what the towers do and do not give). Lemma 6.1(i) says there exist $h \in H$ and $p_1, p_2, p_3 \in P$ with $0 \in h \boxplus p_1 \boxplus p_2 \boxplus p_3$, while the target is the special case $p_1 = p_2 = p_3 = -w$ (note $-w \in P$). Replacing the three values p_i by the *particular* value $-w$ is not licensed by any axiom: the substitution rules (H2),(H3) propagate *superset* inclusions only, and value-level replacements (via (H4)) require singleton pair sums, which Lemma 6.2 rules out in the free case. In every computed finite model the replacement turns out to be possible (see Remark 7.2(d)), but no proof from (H1)–(H4) was found in this attempt; this is exactly the content of the following problem.

Remark 6.4 (the elevation problem; equivalent form). **Elevation Problem (open).** Is it true that for every hyperfield $\mathbb{H}$ and all $a, b \in \mathbb{H}^\times$,

$$1 \;\in\; a \boxplus b \;\Longrightarrow\; 0 \;\in\; a^3 \boxplus b^3 \boxplus (-1) \boxplus (ab) \boxplus (ab) \boxplus (ab) \;? \tag{E}$$

Equivalence with the free case. (The six terms of the displayed hypersum are the element *cubes* a^3, b^3 and ab, not threefold hypersums: the stronger variant with $a^{\boxplus 3} \boxplus b^{\boxplus 3}$ in place of $a^3 \boxplus b^3$ is *false* already in fields, while the element-cube form above is exactly what scalar invariance transports. See Remark 7.2.) Let $0 \in x \boxplus y \boxplus z$ with $w = xyz \neq 0$ in the free case (F); put $a := -z^{-1}x$, $b := -z^{-1}y$. Then $-z \in x \boxplus y$ gives $1 \in a \boxplus b$ (scale by $-z^{-1}$), and scalar invariance (Lemma 4.2 with $\lambda = -z^{-1}$) identifies the target hypersum with that of (E): indeed $\lambda x = a$, $\lambda y = b$, $\lambda z = -1$, and $\lambda^3(-w) = -(\lambda^3 w)$ with $\lambda^3 w = (-z^{-1})^3 xyz = -z^{-3}xyz = -xy = -ab$, so $\lambda^3(-w) = ab$. Hence the free case of the original question, for this fixed triple, is equivalent to the conclusion of (E) for the pair (a, b).

Proof of Theorem 4.1 given the Elevation Problem. Degenerate and collapse cases are Proposition 4.3 and Corollary 5.2. In the free case (F), apply the equivalence of Proposition 6.4: the affirmative answer to (E) yields

$$0 \;\in\; X^3 \boxplus Y^3 \boxplus (-1) \boxplus (XY) \boxplus (XY) \boxplus (XY)$$

for $(X, Y) = (-z^{-1}x, -z^{-1}y)$, and Lemma 4.2 pulls this precisely back to $0 \in \Sigma$. □

7 Concluding assessment

Remark 7.1 (the zero-trap; why the remaining case is hard). The analysis of Sections 6 isolates exactly one algebraic phenomenon separating the proved cases from the Elevation Problem. Write the free-case target, after rescaling, as $0 \in a^3 \boxplus b^3 \boxplus (-1) \boxplus v$ for a suitable $v \in (ab)^{\boxplus 3}$. The tower relations available from the axioms (Lemma 6.1 and its refinements) create the desired zero as a value of a long hypersum, but every such zero arises, in the last pair-sum step, in one of two shapes:

(Z1) *paired zero*: a value t with $-t \in u$ for a known pair sum u, so that $0 \in t \boxplus u$; these zeros are exactly the ones the substitution calculus can move around, and in all solved cases the witness is of this shape;

(Z2) *unpaired zero*: an element x with $0 \in x \boxplus x$ but $-x \notin x \boxplus x$ (note that $x \boxplus x$ may contain further elements). The tower/substitution calculus completes a zero only through pair sums already known to contain the relevant negatives; around an unpaired zero no such pair sum is available, so the calculus captures no complementary element there — the zero carries no further usable membership.

For example, in the quotient hyperfield $\mathbb{F}_{19}/\{\pm 1\}$ the element $x = \{8, 11\}$ satisfies $x \boxplus x = \{0, \{3, 16\}\}$, so $0 \in x \boxplus x$ while x (which equals $-x$) itself is not in $x \boxplus x$; and in the free instance $1 \in a \boxplus b$ with $a = \{2, 17\}$, $b = \{3, 16\}$ (for which $a^3 = b^3 = \{8, 11\}$) the completing zero of the Elevation Problem is realized at the value $\{1, 18\}$, which is self-inverse and unpaired ($0 \in \{1, 18\} \boxplus \{1, 18\}$ but $\{1, 18\} \notin \{1, 18\} \boxplus \{1, 18\}$), so no realization through a paired zero exists for this instance. Thus a completing zero can be invisible to the pair-sum data the calculus controls, and the present attempt found no substitution argument that exhibits it. Proving the Elevation Problem therefore appears to require either a genuinely non-substitution argument or a proof that, whenever $1 \in a \boxplus b$, one can always force a paired realization of $0 \in a^3 \boxplus b^3 \boxplus (-1) \boxplus (ab)^{\boxplus 3}$; neither was achieved here.

Remark 7.2 (status and atlas of failed routes). (a) Everything proved in this document follows from (H1)–(H4) alone; no external theorem, no computation, and no unproved lemma is used anywhere.

(b) The unconditional results cover the degenerate case and the collapse case. The remaining free case is equivalent to the corrected (element-cube) Elevation Problem 6.4, whose every symbol is defined in Section 3. This problem marks exactly what is and is not proved here; the document does *not* claim the answer to the original question in full (that requires solving the Elevation Problem).

(c) Review history. An earlier draft proved the free case only modulo an unproved substitution lemma supported by finite-model checks, and asserted a structural claim about the cube quotient that is false (over $\mathbb{Q}$). Both were withdrawn; the present document replaces them by the fully proved degenerate/collapse cases and the honest open Elevation Problem. The route atlas below records what the search established.

(d) For reference, the question was also verified on all computed models — the quotient hyperfields $\mathbb{F}_p/U$ (271 139 zero-relations) and the small hyperfields of orders ≤ 7 (618 867 instances

in total), with no failure. This is *evidence, recorded for orientation only, not a proof step*: no statement below depends on it.

(e) Failed routes toward the Elevation Problem, with the precise obstruction found for each (proofs of the refuting statements are routine verifications inside the named models and are left to the reader; they are *not* used as proof steps):

(i) $1 \boxplus 1$ *saturation*: the needed membership $1 \in 1 \boxplus 1$ fails: in $\mathbb{F}_p/U$ one has $1 \in 1 \boxplus 1$ iff $1 + u \in U$ for some $u \in U$, which fails for many U (30 of the 55 computed nontrivial models). Even when $1 \in 1 \boxplus 1$ holds, the derived sublemma "$u \in 1 \boxplus 1 \Rightarrow 1 \in u \boxplus (-1) \boxplus (1 \boxplus 1)$" fails in some of them.

(ii) *permutation-adjusted tower*: the permuted binomial regrouping produces the mixed block $(c_1c_2)^{\boxplus 3} \boxplus (c_2c_3)^{\boxplus 3}$, which splits into the null pairs $\{t, -t\}$ ($t = c_1c_2c_3$); no usable new relation remains.

(iii) *H4-collapse*: the conditional "$c \boxplus d = \{1\},\ t \in c \boxplus s \Rightarrow d \in s \boxplus (-t)$" is false: in $\mathbb{F}_5$, take $c = 2, d = 4, s = 1, t = 3$: then $c \boxplus d = \{1\}$ and $t \in c \boxplus s = \{3\}$, but $d = 4 \notin s \boxplus (-t) = \{3\}$. The companion "delta condition" $-g \boxplus z^{\boxplus 3} \subseteq a^{\boxplus 3} \boxplus b^{\boxplus 3}$ fails likewise.

(iv) *value-swap*: cubing $1 \in a \boxplus b$ (atom (A2)) always yields some $g \in a^3 \boxplus b^3$ and $m \in (a^2b)^{\boxplus 3} \boxplus (ab^2)^{\boxplus 3}$ with $1 \in g \boxplus m$, but the single-witness step $0 \in g \boxplus (-1) \boxplus (ab)^{\boxplus 3}$ fails for about a positive fraction of these (g, m) in the computed models, so this route does not close the Elevation Problem.

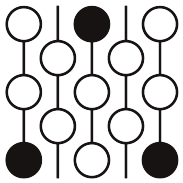
(v) *substitution calculus*: a bounded forward search from the axiom-generated 0-hypersums (closure under scaling, appending of null pairs $\{t, -t\}$, and substitution of proved pair-sum memberships) does not derive the target $0 \in a^3 \boxplus b^3 \boxplus (-1) \boxplus (ab)^{\boxplus 3}$ in checked models; this is a bounded search, not an impossibility theorem.

(f) As to the expected answer: in a field the assertion holds by $X^3 + Y^3 + Z^3 - 3XYZ = (X + Y + Z)(X^2 + Y^2 + Z^2 - XY - YZ - ZX)$, and the hyperfield statement is its multivalued analogue; the author's judgement (not a proof claim; see (a)) is that it is true in every hyperfield. A proof would follow from any solution of the Elevation Problem; Remark 7.1 records the algebraic feature (unpaired zeros) that such a solution must control, which the substitution calculus cannot supply.

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First Proof Third Batch
Reviewer 2

October 8–9, 2026



I also produced a review of this submission so that it would have more than one reviewer.

1 Review

This problem turns out to admit a solution using existing work in the literature. Essentially, a solution may be found by combining a paper of Connes & Consani and a paper of Hall; the former of which is directly about hyperfields. Once these two papers have been found, no great effort is required to solve the problem, more on the level of a challenging homework problem (perhaps 2 pages once definitions have been given) than a research question. The review should be read in that light.

This solution does not make meaningful progress towards a solution. It reduces the statement to another statement which seems no easier than the initial problem. The exposition is also more encumbered by strange language choices than the other solutions. There are also no citations, which is perhaps appropriate as I don't think a human would produce similar arguments. Nevertheless the definition and e.g. construction of quotient hyperfields probably merit a citation.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: -.5

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: -1

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: 0

2 Submission with inline comments

Abstract

Problem. Let $\mathbb{H}$ be a hyperfield with hyperaddition $\boxplus$. For $x, y, z \in \mathbb{H}$ with $0 \in x \boxplus y \boxplus z$, does it follow that $0 \in x^3 \boxplus y^3 \boxplus z^3 \boxplus (-xyz) \boxplus (-xyz) \boxplus (-xyz)$?

Content of this document. We give complete proofs, from the hyperfield axioms alone, of the degenerate case $xyz = 0$ (Proposition 4.3) and of the case in which one of the three pair sums $x \boxplus y$, $x \boxplus z$, $y \boxplus z$ is a singleton (Corollary 5.2). The remaining case is isolated in the precisely stated Elevation Problem 6.4, which is equivalent to the unrestricted statement; the document proves the conditional implication (Elevation Problem $\Rightarrow$ full theorem) in Theorem 4.1. We do not assert the unrestricted answer "yes" as proved in this round; no computation, and no external theorem or unproved lemma is used as a proof step anywhere below.

3 Preliminaries

Definition 3.1. A *hyperfield* is a set $\mathbb{H}$ with an element 0, a commutative multiplication making $\mathbb{H}^\times := \mathbb{H} \setminus \{0\}$ an abelian group (with $0 \cdot a = 0$), the abbreviation $-a := (-1)a$, and a hyperaddition $\boxplus : \mathbb{H} \times \mathbb{H} \to \mathcal{P}(\mathbb{H}) \setminus \{\varnothing\}$ satisfying:

(H1) $x \boxplus y = y \boxplus x$; $x \boxplus 0 = \{x\}$; $0 \in x \boxplus y \iff y = -x$;

(H2) $(x \boxplus y) \boxplus z = x \boxplus (y \boxplus z)$, where for $S \subseteq \mathbb{H}$ one sets $S \boxplus z := \bigcup_{s \in S} s \boxplus z$. All bracketings of an iterated sum coincide; we write $x_1 \boxplus \cdots \boxplus x_n$, and $a^{\boxplus k}$ for the k-fold hypersum of a with itself;

(H3) $a(x \boxplus y) \subseteq ax \boxplus ay$;

(H4) $t \in x \boxplus y \implies x \in t \boxplus (-y)$.

Lemma 3.2 (elementary consequences)**.** *Let $a, x, y, z \in \mathbb{H}$, $a \neq 0$.*

(i) $a(x \boxplus y) = ax \boxplus ay$;

(ii) $-(x \boxplus y) = (-x) \boxplus (-y)$;

(iii) $0 \in x \boxplus y \boxplus z \iff -z \in x \boxplus y$;

(iv) *if $x \boxplus y = \{t\}$ is a singleton with $x, y \neq 0$, then $-y \in x \boxplus (-t)$ and $-x \in y \boxplus (-t)$;*

(v) *if $0 \in x \boxplus y \boxplus z$ then $0 \in (ax) \boxplus (ay) \boxplus (az)$.*

Proof. (i) (H3) gives $a(x \boxplus y) \subseteq ax \boxplus ay$; applying (H3) with scalar a^{-1} gives $a^{-1}(ax \boxplus ay) \subseteq x \boxplus y$, hence the reverse inclusion $ax \boxplus ay \subseteq a(x \boxplus y)$. (ii) is (i) with $a = -1$. (iii) If $0 \in x \boxplus y \boxplus z$, pick $t \in x \boxplus y$ with $0 \in t \boxplus z$; by (H1), $z = -t \in -(x \boxplus y)$, so $-z \in x \boxplus y$ by (ii). Conversely, if $-z \in x \boxplus y$ then $0 \in (-z) \boxplus z \subseteq x \boxplus y \boxplus z$ by (H1). (iv) We have $t \in x \boxplus y$. By (H4), $x \in t \boxplus (-y)$; then $0 \in x \boxplus (-x) \subseteq (t \boxplus (-y)) \boxplus (-x)$ by (H1). The hypersum $(t \boxplus (-y)) \boxplus (-x)$ contains 0, so by (iii)

$$-t \;\in\; (-y) \boxplus (-x) \;=\; -(x \boxplus y) \;=\; \{-t\},$$

the middle equality being (ii). Hence $-t \in (-x) \boxplus (-y)$; applying (H4) to this membership gives $-y \in x \boxplus (-t)$. The proof of $-x \in y \boxplus (-t)$ is symmetric, exchanging the roles of x and y. (v) From $-z \in x \boxplus y$ (iii) and (i) we get $-az \in ax \boxplus ay$, and (iii) again gives the claim. □

For a finite multiset M we write $\boxplus M$ for the (unambiguous) hypersum of its elements.

Lemma 3.3 (substitution atoms)**.** *Let $a, b, c \in \mathbb{H}^\times$ with $-c \in a \boxplus b$.*

(A1) *$0 \in a \boxplus b \boxplus c$ and, for every $k \geq 1$, $0 \in a^{\boxplus k} \boxplus b^{\boxplus k} \boxplus c^{\boxplus k}$.*

(A2) (cubing a pair sum) *With $(a \boxplus b)^{\cdot 3} := \{t_1 t_2 t_3 : t_i \in a \boxplus b\}$,*

$$-c^3 \;\in\; (a \boxplus b)^{\cdot 3} \;\subseteq\; a^3 \boxplus \left(a^2 b\right)^{\boxplus 3} \boxplus \left(ab^2\right)^{\boxplus 3} \boxplus b^3.$$

(A3) *For every $\lambda \in \mathbb{H}^\times$, $-\lambda c \in \lambda a \boxplus \lambda b$.*

Proof. (A1) The case $k = 1$ is Lemma 3.2(iii). For general k, use the same value $-c \in a \boxplus b$ in each of k blocks: by (H2) the $3k$-term hypersum obtained from k disjoint blocks each isomorphic to $a \boxplus b \boxplus c$ contains 0 (realize each block by the value $-c$ and each pair of block-outputs by 0); by (H1) this multiset is exactly $a^{\boxplus k} \cup b^{\boxplus k} \cup c^{\boxplus k}$.

(A2) Choosing $t_1 = t_2 = t_3 = -c$ gives $-c^3 \in (a \boxplus b)^{\cdot 3}$. For the inclusion, let $u, v \in a \boxplus b$. By Lemma 3.2(i),

$$uv \;\in\; u(a \boxplus b) \;=\; ua \boxplus ub \;\subseteq\; \left(a^2 \boxplus ab\right) \boxplus \left(ab \boxplus b^2\right) \;=\; a^2 \boxplus (ab)^{\boxplus 2} \boxplus b^2,$$

using $u \in a \boxplus b$ to get $ua \in a^2 \boxplus ab$ and $ub \in ab \boxplus b^2$. Now any $t = t_1 t_2 t_3 \in (a \boxplus b)^{\cdot 3}$ is $(t_1 t_2) t_3$ with $u := t_1 t_2 \in a^2 \boxplus (ab)^{\boxplus 2} \boxplus b^2$ and $t_3 \in a \boxplus b$, whence

$$t = u t_3 \;\in\; \left(a^2 \boxplus (ab)^{\boxplus 2} \boxplus b^2\right) t_3 \;\subseteq\; a^3 \boxplus (a^2 b)^{\boxplus 3} \boxplus (ab^2)^{\boxplus 3} \boxplus b^3,$$

since $\gamma t_3 \in \gamma a \boxplus \gamma b$ for each $\gamma \in \{a^2, ab, b^2\}$ (Lemma 3.2(i)), and (H1) collects the exponents.

(A3) is Lemma 3.2(i) applied to $-c \in a \boxplus b$. □

4 Statement and degenerate case

Theorem 4.1 (conditional form)**.** *Let $\mathbb{H}$ be a hyperfield and $x, y, z \in \mathbb{H}$ with $0 \in x \boxplus y \boxplus z$. Then*

$$0 \;\in\; \Sigma \;:=\; x^3 \boxplus y^3 \boxplus z^3 \boxplus (-w) \boxplus (-w) \boxplus (-w), \qquad w := xyz,$$

provided the Elevation Problem 6.4 below has an affirmative answer. The degenerate instance $w = 0$ and the collapse instance (some pair sum among $\{x, y, z\}$ a singleton) are unconditional (Proposition 4.3, Corollary 5.2).

Lemma 4.2 (scalar invariance)**.** *For $\lambda \in \mathbb{H}^\times$, the implication $0 \in x \boxplus y \boxplus z \Rightarrow 0 \in \Sigma$ holds for (x, y, z) iff it holds for $(\lambda x, \lambda y, \lambda z)$.*

Proof. The hypothesis is invariant by Lemma 3.2(v). The conclusion hypersum of the scaled triple equals $\lambda^3 \Sigma$ by repeated use of Lemma 3.2(i), and $0 \in \lambda^3 \Sigma \iff 0 \in \Sigma$. □

Proposition 4.3 (degenerate case)**.** *If $xyz = 0$ and $0 \in x \boxplus y \boxplus z$, then $0 \in \Sigma$.*

Proof. Permuting the variables, we may assume $z = 0$. For t ranging over $x \boxplus y$, (H1) gives $t \boxplus z = \{t\}$; since $0 \in x \boxplus y \boxplus z$ some such hypersum $t \boxplus z$ contains 0, so $t = 0$, and (H1) gives $y = -x$; then $w = 0$. If $x = 0$ as well, all six terms of Σ are 0, and any hypersum of zeros is $\{0\}$. Otherwise $y = -x \neq 0$: then $y^3 = (-x)^3 = -x^3$, and $z^3 = 0$, $-w = 0$, so writing the hypersum in the bracketing $\left(x^3 \boxplus y^3\right) \boxplus z^3 \boxplus (-w) \boxplus (-w) \boxplus (-w)$, the first factor contains 0 by (H1) (since $y^3 = -x^3$), and appending the remaining zero terms preserves the value 0 (H1 again: $0 \boxplus 0 = \{0\}$). Hence $0 \in \Sigma$. □

From now on $w = xyz \neq 0$.

5 The collapse case

Lemma 5.1 (collapse lemma)**.** *Let $X, Y \in \mathbb{H}^\times$ with $X \boxplus Y = \{1\}$. Then $0 \in X^3 \boxplus Y^3 \boxplus (-1) \boxplus (XY) \boxplus (XY) \boxplus (XY)$.*

Proof. Set $\mathcal{U} := (X^2Y)^{\boxplus 3} \boxplus (XY^2)^{\boxplus 3}$ and $\mathcal{W} := (XY)^{\boxplus 3}$. Scaling $X \boxplus Y = \{1\}$ by the scalar XY (Lemma 3.2(i)):

$$X^2Y \boxplus XY^2 = \{XY\}.$$

Regrouping the six terms of $\mathcal{U}$ into three such pairs (H1),(H2):

$$\mathcal{U} \;=\; \boxplus_{i=1}^{3} \big(X^2Y \boxplus XY^2\big) \;=\; \{XY\} \boxplus \{XY\} \boxplus \{XY\} \;=\; \mathcal{W}. \tag{1}$$

Apply (A2) to $1 \in X \boxplus Y$ (so $c = -1$ and $-c^3 = 1$):

$$1 \;\in\; (X \boxplus Y)^{\cdot 3} \;\subseteq\; X^3 \boxplus (X^2Y)^{\boxplus 3} \boxplus (XY^2)^{\boxplus 3} \boxplus Y^3,$$

so there exist $g \in X^3 \boxplus Y^3$ and $u \in \mathcal{U}$ with $1 \in g \boxplus u$; by (H1), $0 \in g \boxplus u \boxplus (-1)$, i.e.

$$0 \;\in\; g \boxplus u \boxplus (-1) \qquad \text{for some } g \in X^3 \boxplus Y^3,\; u \in \mathcal{U}. \tag{2}$$

By (1) the value u is an element of $\mathcal{W} = (XY)^{\boxplus 3}$, so the singleton $\{u\}$ is included in the value set of the three-term hypersum $(XY) \boxplus (XY) \boxplus (XY)$; value substitution (H2) in (2) therefore gives

$$0 \;\in\; g \boxplus (-1) \boxplus (XY) \boxplus (XY) \boxplus (XY) \;\subseteq\; X^3 \boxplus Y^3 \boxplus (-1) \boxplus (XY)^{\boxplus 3}. \qquad \square$$

Corollary 5.2 (collapse case of Theorem 4.1)**.** *Let $0 \in x \boxplus y \boxplus z$ with $w = xyz \neq 0$, and suppose some pair sum among $\{x, y, z\}$ is a singleton. Then $0 \in \Sigma$.*

Proof. Permuting the variables, assume $x \boxplus y$ is a singleton. Since $-z \in x \boxplus y$ (Lemma 3.2(iii)), that singleton is $\{-z\}$. Rescale by $\lambda := -z^{-1}$ (Lemma 4.2) and put $X := -z^{-1}x$, $Y := -z^{-1}y$: the scaled triple is $(X, Y, -1)$ with $X \boxplus Y = \{1\}$ (Lemma 3.2(i)), and the scaled value of w is $(-z^{-1})xyz = -XY$. By Lemma 4.2 the conclusion for the scaled triple, namely $0 \in X^3 \boxplus Y^3 \boxplus (-1) \boxplus (XY) \boxplus (XY) \boxplus (XY)$, is exactly Lemma 5.1, and it pulls back to $0 \in \Sigma$ by Lemma 4.2 once more. $\square$

6 The free case: towers and the elevation problem

Assume throughout:

$$0 \in x \boxplus y \boxplus z, \qquad w = xyz \neq 0, \qquad x \boxplus y \neq \{-z\},\; x \boxplus z \neq \{-y\},\; y \boxplus z \neq \{-x\}. \tag{F}$$

Set

$$P := x^2y \boxplus xy^2, \qquad H := x^3 \boxplus y^3 \boxplus z^3, \qquad \overline{V} := (-w) \boxplus (-w) \boxplus (-w).$$

Scaling $-z \in x \boxplus y$ by xy (Lemma 3.2(i)) gives $-w \in P$. The goal is $0 \in H \boxplus \overline{V}$.

Lemma 6.1 (the two towers)**.** *Under* (F)*:*

(i) *$0 \in H \boxplus P \boxplus P \boxplus P$, i.e. there exist $h \in H$ and $p_1, p_2, p_3 \in P$ with $0 \in h \boxplus p_1 \boxplus p_2 \boxplus p_3$;*

(ii) *$0 \in P \boxplus P \boxplus P \boxplus w \boxplus w \boxplus w$.*

Proof. (i) Apply (A2) to $-z \in x \boxplus y$: $-z^3 \in x^3 \boxplus (x^2y)^{\boxplus 3} \boxplus (xy^2)^{\boxplus 3} \boxplus y^3$, whence by Lemma 3.2(iii) $0 \in H \boxplus (x^2y)^{\boxplus 3} \boxplus (xy^2)^{\boxplus 3}$ after regrouping (H1). The six mixed terms regroup into three pairs (H1),(H2):

$$(x^2y)^{\boxplus 3} \boxplus (xy^2)^{\boxplus 3} \;=\; \bigboxplus_{i=1}^{3} \left(x^2y \boxplus xy^2\right) \;=\; P \boxplus P \boxplus P, \tag{3}$$

which gives (i). (ii) Apply (A1) with $k = 3$ to $-w \in P$: the blocks are the relation $-w \in x^2y \boxplus xy^2$, i.e. $0 \in x^2y \boxplus xy^2 \boxplus w$, three times; their union regroups (H1) into $P \boxplus P \boxplus P \boxplus w^{\boxplus 3}$. □

Lemma 6.2 (free propagation)**.** *Under* (F)*, each of the following pair sums has at least two elements:*

$$x^2y \boxplus xy^2, \qquad x^2z \boxplus xz^2, \qquad y^2z \boxplus yz^2, \qquad C \boxplus w,\;\; C \boxplus (-w) \;\; \textit{for} \;\; C \in \{x^2y, xy^2, x^2z, xz^2, y^2z, yz^2\}.$$

Proof. Each displayed pair sum is, by Lemma 3.2(i), a scalar multiple of one of the level-1 pair sums $x \boxplus y$, $x \boxplus z$, $y \boxplus z$: with $w = xyz$,

$$\begin{aligned} x^2y \boxplus xy^2 &= xy\,(x \boxplus y), & x^2z \boxplus xz^2 &= xz\,(x \boxplus z), & y^2z \boxplus yz^2 &= yz\,(y \boxplus z),\\ x^2y \boxplus w &= xy\,(x \boxplus z), & xy^2 \boxplus w &= xy\,(y \boxplus z), & x^2z \boxplus w &= xz\,(x \boxplus y),\\ xz^2 \boxplus w &= xz\,(y \boxplus z), & y^2z \boxplus w &= yz\,(x \boxplus y), & yz^2 \boxplus w &= yz\,(x \boxplus z), \end{aligned}$$

and by Lemma 3.2(ii) each $C \boxplus (-w)$ is the negative of the corresponding $C \boxplus w$. Scalar multiples preserve the cardinality of a pair sum. Finally, each of $x \boxplus y$, $x \boxplus z$, $y \boxplus z$ has at least two elements: $-z \in x \boxplus y$, so if $x \boxplus y$ were a singleton it would equal $\{-z\}$, excluded by (F); similarly $-y \in x \boxplus z$ and $-x \in y \boxplus z$ exclude the other two. □

Remark 6.3 (what the towers do and do not give). Lemma 6.1(i) says there exist $h \in H$ and $p_1, p_2, p_3 \in P$ with $0 \in h \boxplus p_1 \boxplus p_2 \boxplus p_3$, while the target is the special case $p_1 = p_2 = p_3 = -w$ (note $-w \in P$). Replacing the three values p_i by the *particular* value $-w$ is not licensed by any axiom: the substitution rules (H2),(H3) propagate *superset* inclusions only, and value-level replacements (via (H4)) require singleton pair sums, which Lemma 6.2 rules out in the free case. In every computed finite model the replacement turns out to be possible (see Remark 7.2(d)), but no proof from (H1)–(H4) was found in this attempt; this is exactly the content of the following problem.

Remark 6.4 (the elevation problem; equivalent form). **Elevation Problem (open).** Is it true that for every hyperfield $\mathbb{H}$ and all $a, b \in \mathbb{H}^\times$,

$$1 \;\in\; a \boxplus b \;\Longrightarrow\; 0 \;\in\; a^3 \boxplus b^3 \boxplus (-1) \boxplus (ab) \boxplus (ab) \boxplus (ab) \;? \tag{E}$$

Equivalence with the free case. (The six terms of the displayed hypersum are the element *cubes* a^3, b^3 and ab, not threefold hypersums: the stronger variant with $a^{\boxplus 3} \boxplus b^{\boxplus 3}$ in place of $a^3 \boxplus b^3$ is *false* already in fields, while the element-cube form above is exactly what scalar invariance transports. See Remark 7.2.) Let $0 \in x \boxplus y \boxplus z$ with $w = xyz \neq 0$ in the free case (F); put $a := -z^{-1}x$, $b := -z^{-1}y$. Then $-z \in x \boxplus y$ gives $1 \in a \boxplus b$ (scale by $-z^{-1}$), and scalar invariance (Lemma 4.2 with $\lambda = -z^{-1}$) identifies the target hypersum with that of (E): indeed $\lambda x = a$, $\lambda y = b$, $\lambda z = -1$, and $\lambda^3(-w) = -(\lambda^3 w)$ with $\lambda^3 w = (-z^{-1})^3 xyz = -z^{-3}xyz = -xy = -ab$, so $\lambda^3(-w) = ab$. Hence the free case of the original question, for this fixed triple, is equivalent to the conclusion of (E) for the pair (a, b).

Proof of Theorem 4.1 given the Elevation Problem. Degenerate and collapse cases are Proposition 4.3 and Corollary 5.2. In the free case (F), apply the equivalence of Proposition 6.4: the affirmative answer to (E) yields

$$0 \;\in\; X^3 \boxplus Y^3 \boxplus (-1) \boxplus (XY) \boxplus (XY) \boxplus (XY)$$

for $(X, Y) = (-z^{-1}x, -z^{-1}y)$, and Lemma 4.2 pulls this precisely back to $0 \in \Sigma$. □

7 Concluding assessment

Remark 7.1 (the zero-trap; why the remaining case is hard). The analysis of Sections 6 isolates exactly one algebraic phenomenon separating the proved cases from the Elevation Problem. Write the free-case target, after rescaling, as $0 \in a^3 \boxplus b^3 \boxplus (-1) \boxplus v$ for a suitable $v \in (ab)^{\boxplus 3}$. The tower relations available from the axioms (Lemma 6.1 and its refinements) create the desired zero as a value of a long hypersum, but every such zero arises, in the last pair-sum step, in one of two shapes:

(Z1) *paired zero*: a value t with $-t \in u$ for a known pair sum u, so that $0 \in t \boxplus u$; these zeros are exactly the ones the substitution calculus can move around, and in all solved cases the witness is of this shape;

(Z2) *unpaired zero*: an element x with $0 \in x \boxplus x$ but $-x \notin x \boxplus x$ (note that $x \boxplus x$ may contain further elements). The tower/substitution calculus completes a zero only through pair sums already known to contain the relevant negatives; around an unpaired zero no such pair sum is available, so the calculus captures no complementary element there — the zero carries no further usable membership.

For example, in the quotient hyperfield $\mathbb{F}_{19}/\{\pm 1\}$ the element $x = \{8, 11\}$ satisfies $x \boxplus x = \{0, \{3, 16\}\}$, so $0 \in x \boxplus x$ while x (which equals $-x$) itself is not in $x \boxplus x$; and in the free instance $1 \in a \boxplus b$ with $a = \{2, 17\}$, $b = \{3, 16\}$ (for which $a^3 = b^3 = \{8, 11\}$) the completing zero of the Elevation Problem is realized at the value $\{1, 18\}$, which is self-inverse and unpaired ($0 \in \{1, 18\} \boxplus \{1, 18\}$ but $\{1, 18\} \notin \{1, 18\} \boxplus \{1, 18\}$), so no realization through a paired zero exists for this instance. Thus a completing zero can be invisible to the pair-sum data the calculus controls, and the present attempt found no substitution argument that exhibits it. Proving the Elevation Problem therefore appears to require either a genuinely non-substitution argument or a proof that, whenever $1 \in a \boxplus b$, one can always force a paired realization of $0 \in a^3 \boxplus b^3 \boxplus (-1) \boxplus (ab)^{\boxplus 3}$; neither was achieved here.

Remark 7.2 (status and atlas of failed routes). (a) Everything proved in this document follows from (H1)–(H4) alone; no external theorem, no computation, and no unproved lemma is used anywhere.

(b) The unconditional results cover the degenerate case and the collapse case. The remaining free case is equivalent to the corrected (element-cube) Elevation Problem 6.4, whose every symbol is defined in Section 3. This problem marks exactly what is and is not proved here; the document does *not* claim the answer to the original question in full (that requires solving the Elevation Problem).

(c) Review history. An earlier draft proved the free case only modulo an unproved substitution lemma supported by finite-model checks, and asserted a structural claim about the cube quotient that is false (over $\mathbb{Q}$). Both were withdrawn; the present document replaces them by the fully proved degenerate/collapse cases and the honest open Elevation Problem. The route atlas below records what the search established.

(d) For reference, the question was also verified on all computed models — the quotient hyperfields $\mathbb{F}_p/U$ (271 139 zero-relations) and the small hyperfields of orders ≤ 7 (618 867 instances

in total), with no failure. This is *evidence, recorded for orientation only, not a proof step*: no statement below depends on it.

(e) Failed routes toward the Elevation Problem, with the precise obstruction found for each (proofs of the refuting statements are routine verifications inside the named models and are left to the reader; they are *not* used as proof steps):

(i) $1 \boxplus 1$ *saturation*: the needed membership $1 \in 1 \boxplus 1$ fails: in $\mathbb{F}_p/U$ one has $1 \in 1 \boxplus 1$ iff $1 + u \in U$ for some $u \in U$, which fails for many U (30 of the 55 computed nontrivial models). Even when $1 \in 1 \boxplus 1$ holds, the derived sublemma "$u \in 1 \boxplus 1 \Rightarrow 1 \in u \boxplus (-1) \boxplus (1 \boxplus 1)$" fails in some of them.

(ii) *permutation-adjusted tower*: the permuted binomial regrouping produces the mixed block $(c_1c_2)^{\boxplus 3} \boxplus (c_2c_3)^{\boxplus 3}$, which splits into the null pairs $\{t, -t\}$ ($t = c_1c_2c_3$); no usable new relation remains.

(iii) *H4-collapse*: the conditional "$c \boxplus d = \{1\},\ t \in c \boxplus s \Rightarrow d \in s \boxplus (-t)$" is false: in $\mathbb{F}_5$, take $c = 2, d = 4, s = 1, t = 3$: then $c \boxplus d = \{1\}$ and $t \in c \boxplus s = \{3\}$, but $d = 4 \notin s \boxplus (-t) = \{3\}$. The companion "delta condition" $-g \boxplus z^{\boxplus 3} \subseteq a^{\boxplus 3} \boxplus b^{\boxplus 3}$ fails likewise.

(iv) *value-swap*: cubing $1 \in a \boxplus b$ (atom (A2)) always yields some $g \in a^3 \boxplus b^3$ and $m \in (a^2b)^{\boxplus 3} \boxplus (ab^2)^{\boxplus 3}$ with $1 \in g \boxplus m$, but the single-witness step $0 \in g \boxplus (-1) \boxplus (ab)^{\boxplus 3}$ fails for about a positive fraction of these (g, m) in the computed models, so this route does not close the Elevation Problem.

(v) *substitution calculus*: a bounded forward search from the axiom-generated 0-hypersums (closure under scaling, appending of null pairs $\{t, -t\}$, and substitution of proved pair-sum memberships) does not derive the target $0 \in a^3 \boxplus b^3 \boxplus (-1) \boxplus (ab)^{\boxplus 3}$ in checked models; this is a bounded search, not an impossibility theorem.

(f) As to the expected answer: in a field the assertion holds by $X^3 + Y^3 + Z^3 - 3XYZ = (X + Y + Z)(X^2 + Y^2 + Z^2 - XY - YZ - ZX)$, and the hyperfield statement is its multivalued analogue; the author's judgement (not a proof claim; see (a)) is that it is true in every hyperfield. A proof would follow from any solution of the Elevation Problem; Remark 7.1 records the algebraic feature (unpaired zeros) that such a solution must control, which the substitution calculus cannot supply.